

Parker Whitfill

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Summary

Economics PhD researcher with deep expertise in AI/ML systems, benchmarking, and computational methods. Published in *Nature* on large-scale AI evaluation. Research collaborator with Epoch AI and METR on AI forecasting, compute economics, and algorithmic progress. Experienced in building ML pipelines for OCR, developing AI benchmarks, and full-stack software development. Proficient in Python, R, and modern ML frameworks.

Education

Ph.D. in Economics 2023 – Present
Massachusetts Institute of Technology Cambridge, MA

- Research focus: AI capabilities evaluation, benchmark design, compute economics, algorithmic progress
- Coursework in machine learning, econometrics, causal inference, and computational methods

B.A. in Economics 2017 – 2021
University of Chicago Chicago, IL

Research & Professional Experience

Future of Life Institute Fellow – Software Engineer 2025
Future of Life Institute Cambridge, MA

- Designed and built a cooperative technology platform enabling users to coordinate messaging and collective action, from architecture through deployment
- Developed full-stack application using Python and JavaScript, integrating APIs for real-time communication and user coordination workflows
- Implemented backend services, database schema design, and frontend interfaces over a 3-month intensive fellowship focused on AI governance and safety tooling

Research Assistant 2021 – 2023
University of Chicago Chicago, IL

- Designed and implemented machine learning pipelines for optical character recognition (OCR) of historical documents, automating extraction of structured data from large archival corpora
- Built text preprocessing, image segmentation, and classification models using Python (PyTorch, OpenCV, Tesseract) to digitize and parse historical economic records
- Developed data ingestion and cleaning workflows for large-scale historical datasets, enabling quantitative analysis of inequality and institutional change
- Conducted empirical research on economic history, contributing to a publication on the causal effects of WWII on inequality and the social contract in Britain
- Managed end-to-end research pipelines: data collection, feature engineering, statistical modeling, and visualization

Publications & Research

A benchmark of expert-level academic questions to assess AI capabilities

Coauthor; consortium paper with >1,000 authors

Nature, 649:1139–1146, 2026. [doi:10.1038/s41586-025-09962-4](https://doi.org/10.1038/s41586-025-09962-4)

- Co-developed Humanity’s Last Exam (HLE), a 3,000-question multi-modal benchmark evaluating frontier AI systems across dozens of academic disciplines
- Contributed to benchmark design, question curation, and evaluation methodology for assessing AI capabilities at expert level

Do Benchmarks Overstate AI Capabilities? Evidence from SWE-Bench Verified

Parker Whitfill, Cheryl Wu, Nate Rush, Joel Becker

Working Paper

- Investigating measurement validity of widely-used software engineering AI benchmarks

Will Compute Bottlenecks Prevent an Intelligence Explosion?

Parker Whitfill and Cheryl Wu. [arXiv:2507.23181](https://arxiv.org/abs/2507.23181)

- Modeled compute supply-demand dynamics for AI scaling; estimated substitutability between compute and algorithmic efficiency

The Software Intelligence Explosion Debate Needs Experiments

Anson Ho (Epoch AI), Parker Whitfill. [Epoch AI](https://epoch.ai)

- Collaboration with Epoch AI analyzing feasibility of recursive AI self-improvement; proposed experimental frameworks for measuring AI R&D productivity

Forecasting AI Time Horizon Under Compute Slowdowns

Parker Whitfill, Ben Snodin, Joel Becker (METR). [arXiv:2511.19492](https://arxiv.org/abs/2511.19492)

- Collaboration with METR; built quantitative forecasting models for AI development timelines under hardware constraint scenarios

Note on Selection Bias in Observational Estimates of Algorithmic Progress

Parker Whitfill. [arXiv:2508.11033](https://arxiv.org/abs/2508.11033)

Beyond Ordinal Preferences: Why Alignment Needs Cardinal Human Feedback

Parker Whitfill and Stewart Slocum. [arXiv:2508.08486](https://arxiv.org/abs/2508.08486)

- Analyzed limitations of RLHF preference models; proposed cardinal feedback mechanisms for AI alignment

The Second World War, Inequality and the Social Contract in England

Leander Heldring, James Robinson, Parker Whitfill (equal contribution)

Economica, Volume 89, Issue S1, Pages S137–S159

Technical Skills

Languages: Python, R, SQL, $\text{\LaTeX}$, Bash, JavaScript/HTML/CSS

ML/AI: PyTorch, scikit-learn, OpenCV, Tesseract OCR, Hugging Face Transformers, LLM evaluation

Data & Compute: Pandas, NumPy, Stata, large-scale data processing, statistical modeling

Tools: Git, Linux/Unix, cloud computing, Jupyter, VS Code

Methods: Machine learning, deep learning, computer vision (OCR), NLP, causal inference, econometrics, benchmark design & evaluation, time-series forecasting

Honors & Fellowships

- Emergent Ventures Grantee (2025)
- Stripe Economics of AI Fellowship (2025)
- Future of Life Institute Fellowship (2025)
- 2017 National Debate Champion (TOC LD)